

Drug Repurposing

Drug Repurposing for Asthma (ICD-10 J45)

Aim: The aim of this practical assignment was to develop and evaluate a machine learning-based framework for identifying compounds with potential repurposing value for asthma treatment. Specifically, the main objectives involve: to transform molecular compound data into machine-readable descriptors, to construct predictive classification models capable of distinguishing active and inactive compounds, to assess the dataset imbalance impact on model performance, evaluation of balancing strategies and parameter optimisation, and to determine the practical applicability of the resulting predictive model for drug repurposing.

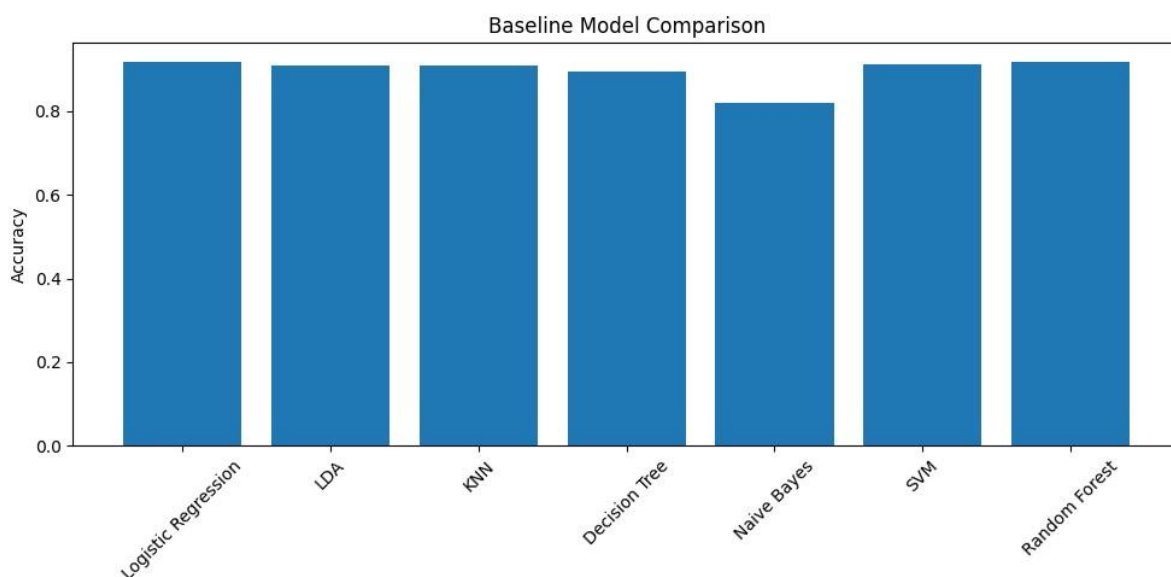
Task Description: This assignment involved the construction of a supervised machine learning pipeline using a dataset containing molecular compounds represented by Simplified Molecular Input Line Entry System (SMILES) strings and their associated ICD-10 disease classification codes. The objective is to identify compounds that are potentially active against asthma (ICD-10 code J45). The required tasks included preprocessing the dataset, generating molecular descriptors, constructing the input feature matrix and binary target labels, analysing class distribution, training multiple machine learning models, evaluating the effect of balancing techniques such as oversampling and undersampling, testing parameter modifications, and critically assessing the predictive utility and limitations of the final model.

Method: The dataset was imported into Python using pandas and cleaned to extract molecular structures and disease labels. SMILES strings were converted into numerical molecular descriptors using RDKit-generated Morgan fingerprints (radius = 2, 1024 bits) that are commonly used in drug discovery to capture structural features efficiently. Asthma-related compounds were identified using the broader ICD-10 respiratory disease category (J40–J47) that contains relevant compounds labelled as active and all others as inactive. Invalid molecular structures were removed during preprocessing. The class distribution was then examined to assess dataset balance, followed by a stratified train–test split to preserve class proportions. Seven machine learning models were tested, including Logistic Regression, LDA, KNN, Decision Tree, Gaussian Naïve Bayes, SVM, and Random Forest and stratified 10-fold cross-validation with accuracy as the main evaluation metric.

To address class imbalance, two sampling strategies were implemented. Random oversampling was used to duplicate minority-class examples until both classes were equally represented and random undersampling reduced the majority class to match the minority class size. The same classification models were retrained using these balanced datasets to assess the impact of sampling strategies. To investigate whether further improvements could be achieved, hyperparameter tuning was performed on selected models. The important parameter adjustment such as Random Forest tree depth was modified and SVM kernel configuration was modified, and the number of neighbours in KNN was adjusted. Finally, the best-performing model was evaluated using a

heldout test dataset and its performance was assessed using accuracy, confusion matrix analysis, precision, recall, and F1-score.

Results and Discussion: The preprocessing stage successfully transformed the raw molecular dataset into a structured machine learning format suitable for predictive modelling. Molecular fingerprints preserved relevant structural chemical information and leads to the application of supervised learning techniques. Class distribution analysis revealed substantial imbalance in the dataset. The datasets comprises of 998 compounds labelled as inactive, while only 107 compounds were classified as active that leads to an approximate imbalance ratio of 9.3:1. This imbalance represented a major methodological challenge because classifiers trained on such skewed data often develop bias toward the majority class. In the context of drug repurposing, this is problematic because false negatives may result in potentially useful therapeutic candidates being overlooked. **Fig 7: Baseline model comparison of all models**



Baseline model performance showed relatively high accuracy, with Logistic Regression and Random Forest achieving around 92% and most other models performed close to 91%, except Naïve Bayes at 82%. These results indicates all the models have predictive efficiency and potential. However, these results should be interpreted cautiously due to class imbalance as accuracy may reflect bias toward the majority class.

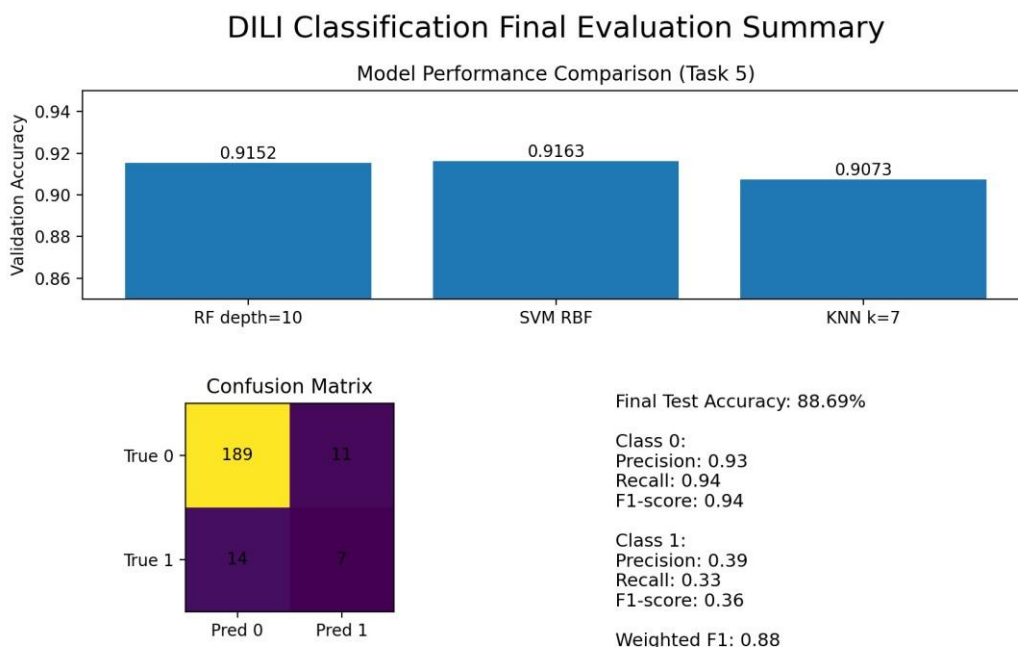
Table 7: Model performance after over and under sampling

MODEL / ORIGINAL DATASET OVERSAMPLED UNDERSAMPLED DATASET (CLASS DISTRIBUTION) ACCURACY ACCURACY			
DATASET DISTRIBUTION	[798, 86]	[798, 798]	[86, 86]
LOGISTIC REGRESSION	—	0.9530	0.7092

LDA	—	0.9449	0.7095
KNN	—	0.9455	0.5637
DECISION TREE	—	0.9392	0.6631
NAIVE BAYES	—	0.9367	0.6399
SVM	—	0.9599	0.6745
RANDOM FOREST	—	0.9593	0.6863

Applying random oversampling significantly improved performance, with SVM achieving the highest accuracy (95.99%) and followed closely by Random Forest (95.93%) indicating that balancing the dataset improved the models' ability to learn minority-class patterns. In contrast, random undersampling reduced performance considerably across all models, likely due to information loss from removing many majority-class samples.

Fig 8: Model performance comparison (Hyperparameter tuning)



Hyperparameter tuning did not improve results beyond the oversampled models. Final testing of the oversampled Random Forest model achieved 88.69% accuracy and performance for identifying active asthma-related compounds remained weak with low precision (0.39), recall (0.33), and F1-score (0.36). This suggests that although overall accuracy was high, the model struggled to detect the minority class effectively with the gap between cross-validation and test performance indicating possible overfitting caused by oversampling.

Discussion

This study successfully developed a machine learning-based drug repurposing framework for asthma using molecular fingerprint descriptors and supervised classification methods. The findings demonstrate that machine learning can effectively extract predictive signals from

molecular structural data and achieve strong apparent classification accuracy. Balancing strategies, particularly oversampling, significantly improved model performance by reducing class imbalance bias. However, the practical utility of the final model remains limited due to poor sensitivity toward active compounds. The low recall for asthma-associated compounds indicates that a substantial proportion of potentially valuable therapeutic candidates would be missed that is not an ideal situation in real-world drug discovery applications where identifying promising compounds is a priority.

Several limitations must be addressed before practical implementation. First, the dataset relied on ICD-10 level 2 disease classifications rather than direct asthma-specific biological activity data, introducing label ambiguity. Second, accuracy alone proved insufficient for evaluating performance in an imbalanced biomedical context that indicates the importance of recall, F1-score, ROC-AUC, and precision-recall analysis. Third, Morgan fingerprints represent only one form of molecular descriptor, and alternative representations such as MACCS fingerprints, physicochemical descriptors, or graph neural network embeddings could also improve predictive performance. Future work should include more advanced feature engineering, systematic hyperparameter optimisation, use of alternative balancing strategies such as SMOTE, evaluation on independent external datasets, molecular docking studies, pathway analysis, laboratory biological validation, toxicity assessment, and clinical evaluation.